CoRoT-2b

R-0218

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_CoRoT-2_180912` · created 2026-07-17T01:03:52+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2458373.6717 +/- 0.0015 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.158 +/- 0.039
Transit depth (Rp/Rs)^2	2.5 +/- 1.23 [%]
Orbital Inclination (inc)	89.99 +/- 1.71
Airmass coefficient 1 (a1)	1.01 +/- 0.015
Airmass coefficient 2 (a2)	-0.009 +/- 0.055
Scatter in the residuals of the lightcurve fit is	1.46 %
Best Comparison Star	#2 - [564, 130]
Optimal Aperture	3.38
Optimal Annulus	9.33
Transit Duration (day)	0.0935 +/- 0.003

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.158 ± 0.039 vs 0.1667 (deviation 0.2σ)
Duration fit vs published	0.0935 d vs 0.095 d (deviation 0.45σ)
Mid-time O–C	4.8 min
Depth SNR	3.6
Residual scatter	1.46 %

Light curve

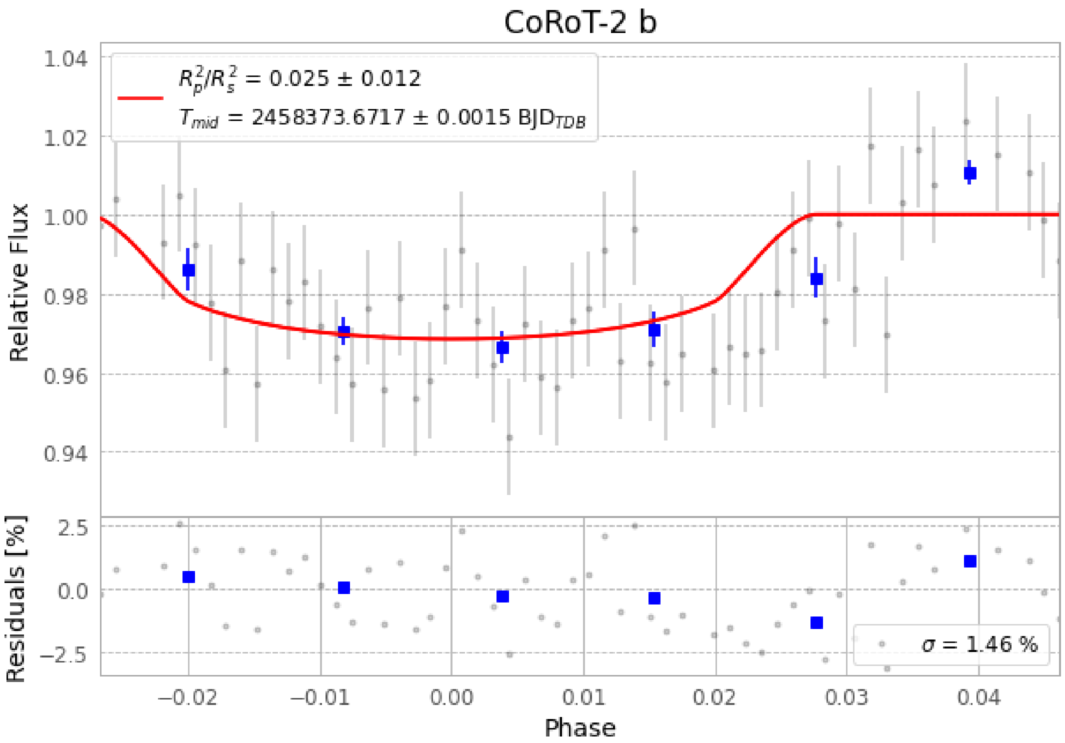

Figure 1 — FinalLightCurve_CoRoT-2 b_2018-09-11.png (SHA-256 ee04c3d79fd28c3f…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [274, 238], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 e5eeaf655cfdbf44…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 825eb7e`

Data provenance

59 FITS frames (39 MB), `CoRoT-2180912025412.FITS` → `CoRoT-2180912055712.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-CoRoT-2-180912.log (f4e03f3faceb…), FinalLightCurve_CoRoT-2 b_2018-09-11.png (ee04c3d79fd2…), FinalLightCurve_CoRoT-2 b_2018-09-11.csv (ad4b60fb6c59…)

Compute resources

Wall time 105.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)